Theoretical ecotoxicology

Multiple stressors: Stressor interactions

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Iris Madge Pimentel & Ralf Schäfer

My research

- PostDoc in Leese Lab – Aquatic ecosystem research
- Multiple stressor effects on macroinvertebrate communities
  - DNA-based methods
  - Mesocosm experiments
  - Theoretical work on how multiple stressors combine

AQUATISCHE ÖKOSYSTEM FORSCHUNG

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Learning outcomes of today's lecture

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1. **Understanding toxicant mixtures and how their toxicity can be estimated**
   - Something-from-nothing effect
2. **Which mixture models have good predictive power under which conditions?**
   - Independent action model
   - Concentration addition model
   - When to choose which model?
   - How well to the models match observations and how strong is the deviation?
3. **Combining environmental stressors with toxicants: Multiple-stressor null models**
   - Understanding synergism and antagonism and why null-model choice matters
   - Applying criteria for null-model choice

Chemicals do not occur in isolation!

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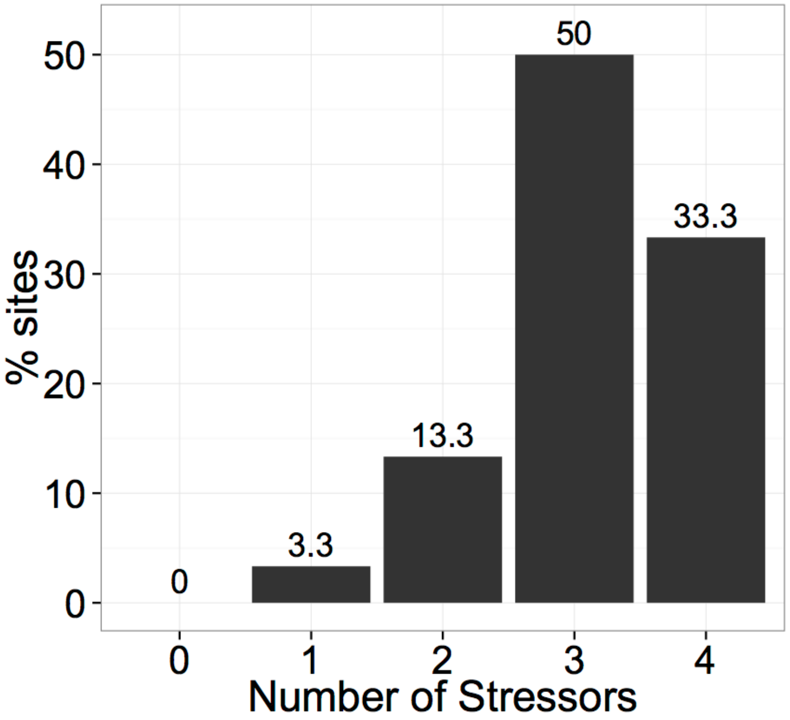
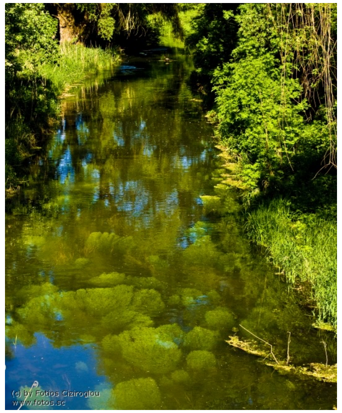
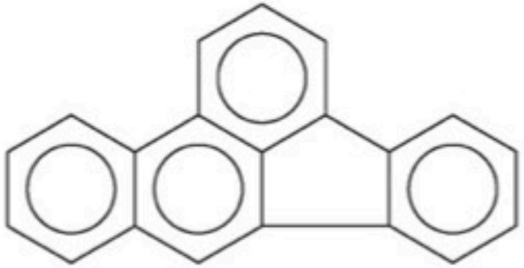
Analysis of german WFD monitoring data



Schäfer et al. 2016 *Freshw. Biol.* 61: 2116



Excessive nutrients

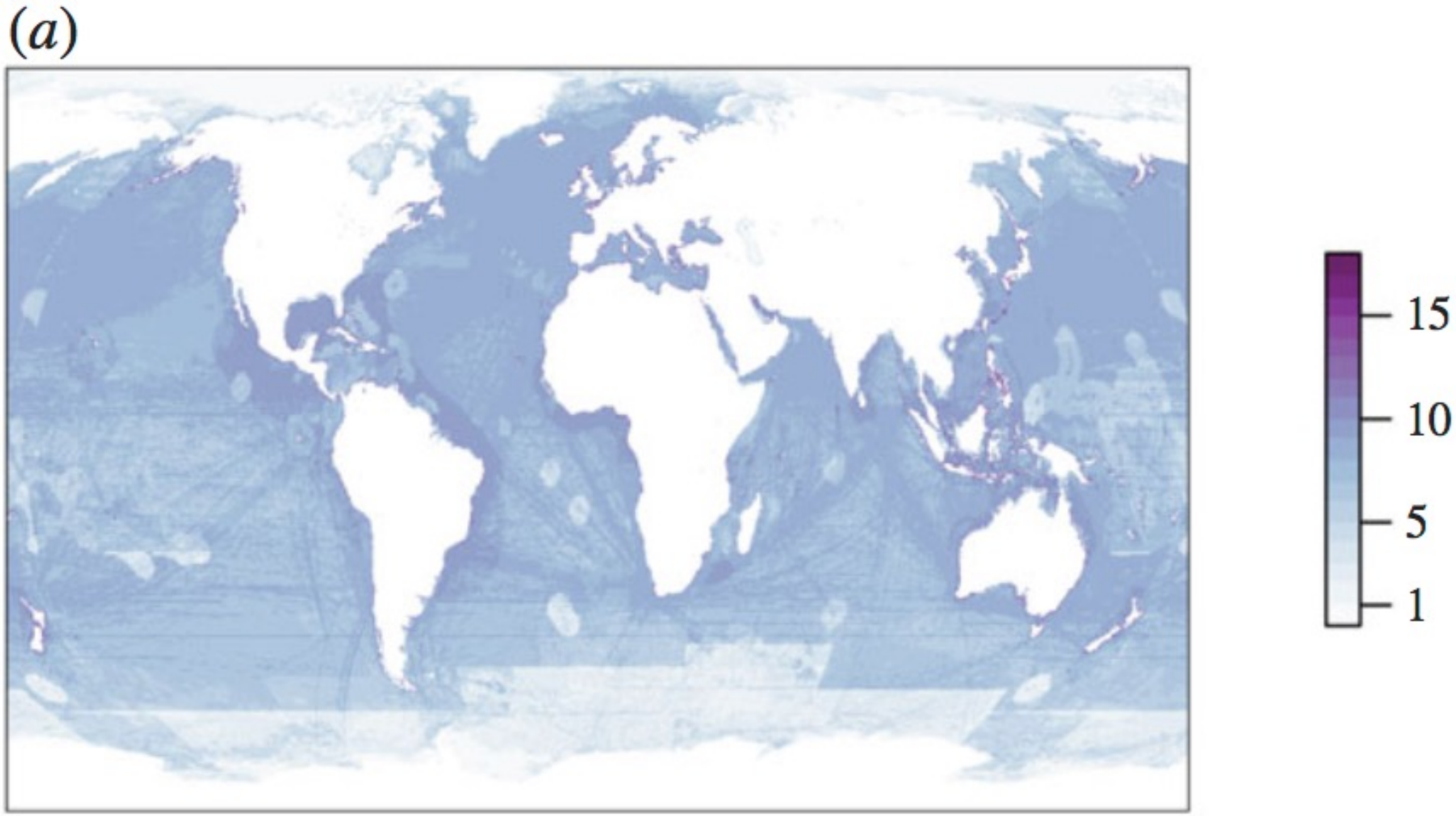


Toxicants

Invasive species

Habitat degradation

Marine ecosystems: Most areas experience many stressors

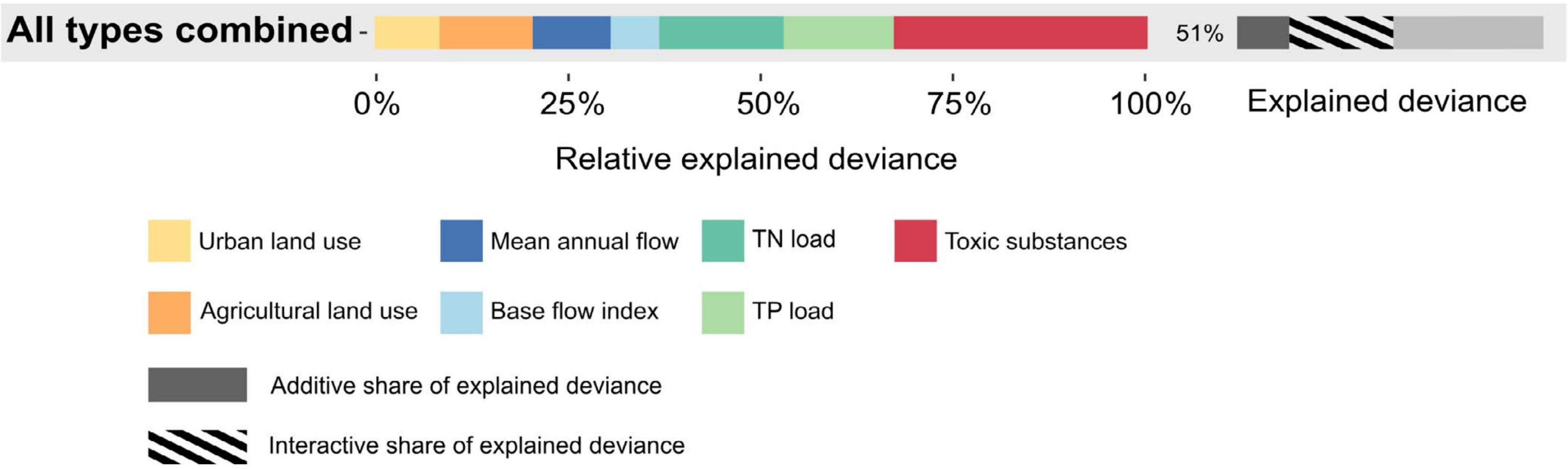


Côté et al. 2016 *Proc. Roy. Soc. B* 283: 20152592

Several (drivers of) stressors explain variability in ecological status

Analysis of European WFD monitoring data

Interactions among stressors play a huge role!



Lemm et al. 2021 *Glob. Ch. Biol. 27*: 1962

A wide range of chemical mixtures with different modes of action and multiple non-chemical stressors co-occur that can potentially affect organisms and may interact

Mechanisms of stressor interactions

Intensity interactions

= Changes in stressor intensity (or exposure) by another stressor

Sediment addition

Reduces →

Predation by predatory stoneflies (Perlidae)

antagonism

Louhi et al. 2017
Can. J. Fish. Aqu. Sci. 74: 32

Pics from wikimedia and macroinvertebrates.org

Mechanisms of stressor interactions

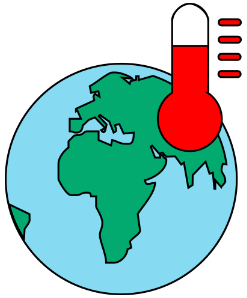
Effect interactions

= Changes in *per-capita* stressor effects by another stressor, e.g. by modifying an organism's sensitivity

Food limitation

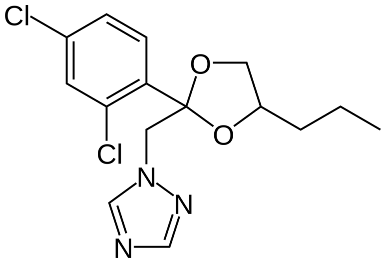
Orcutt & Porter 1984, *Oecologia* 63: 300

Increases sensitivity to…



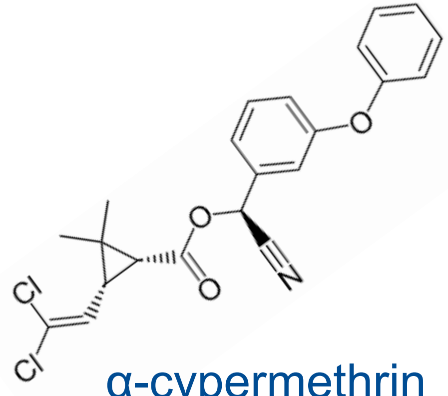
Warming

Propicon-azole



Cedergreen et al. 2017 *ES&T* 51:14379

Inhibits biotransformation of…



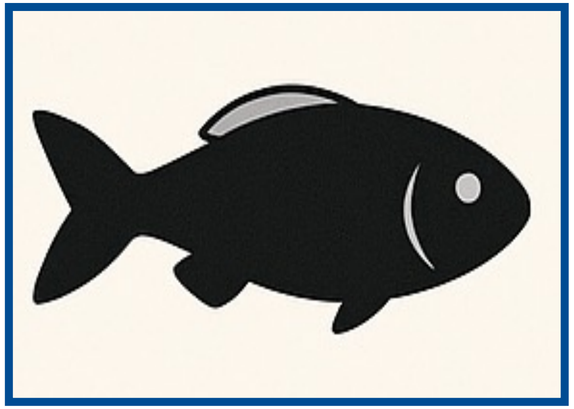
α-cypermethrin

synergism

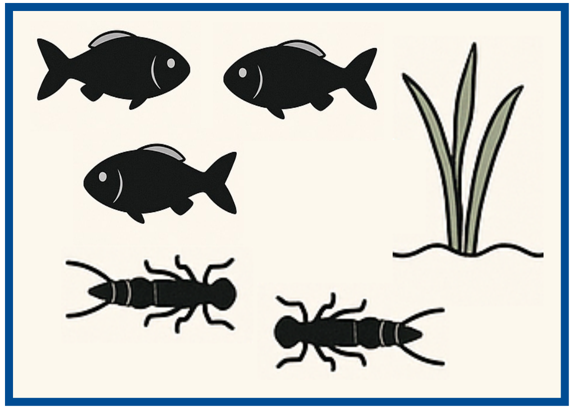
Daphnia sp.

Where do stressor interactions occur?

Abiotic environment
- Purely physical/chemical processes
- E.g., drought increasing contaminant concentrations in stream

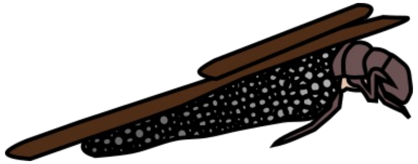


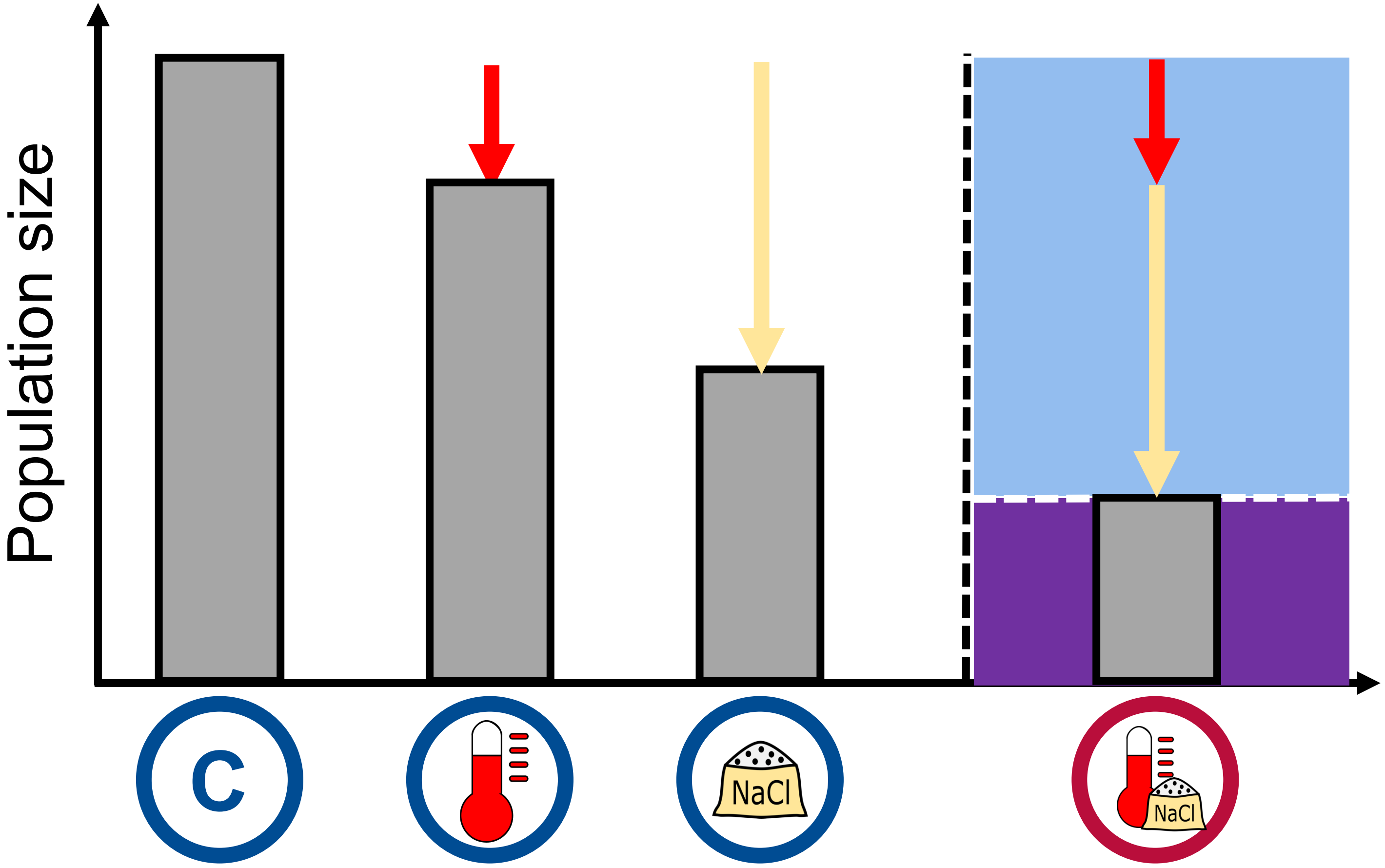
Organismal level
- Through physiological processes within an organism
- E.g., the inhibition of biotransformation in the example before



Ecosystem level
- Alterations of biotic interactions
- E.g., through changes in predation pressure (stonefly example)

Interactions as deviation from null-model predictions





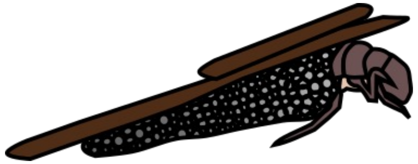
Antagonism

Additive (null model)

Synergism

Multiple-stressor null models

Multiple-stressor null model:
A model that predicts the joint effect assuming the absence of interactions



C — C = 100

$S_A = 80$ — -20, x80%

NaCl — $S_B = 50$ — -50, x50%

Null models for stressor co-occurrence

Simple addition:
$S_{A,B} = S_A + S_B - C$

NaCl — -70 — 30

Multiplicative:
$S_{A,B} = (S_A \times S_B)/C$

NaCl — x40% — 40

Dominance:
$\min(S_{A,B}, C) = \min(S_A, S_B)$

NaCl — min(80, 50) — 50

Madge Pimentel et al. (2025). In: *Ecology & Evolution*

Which null model should I choose?

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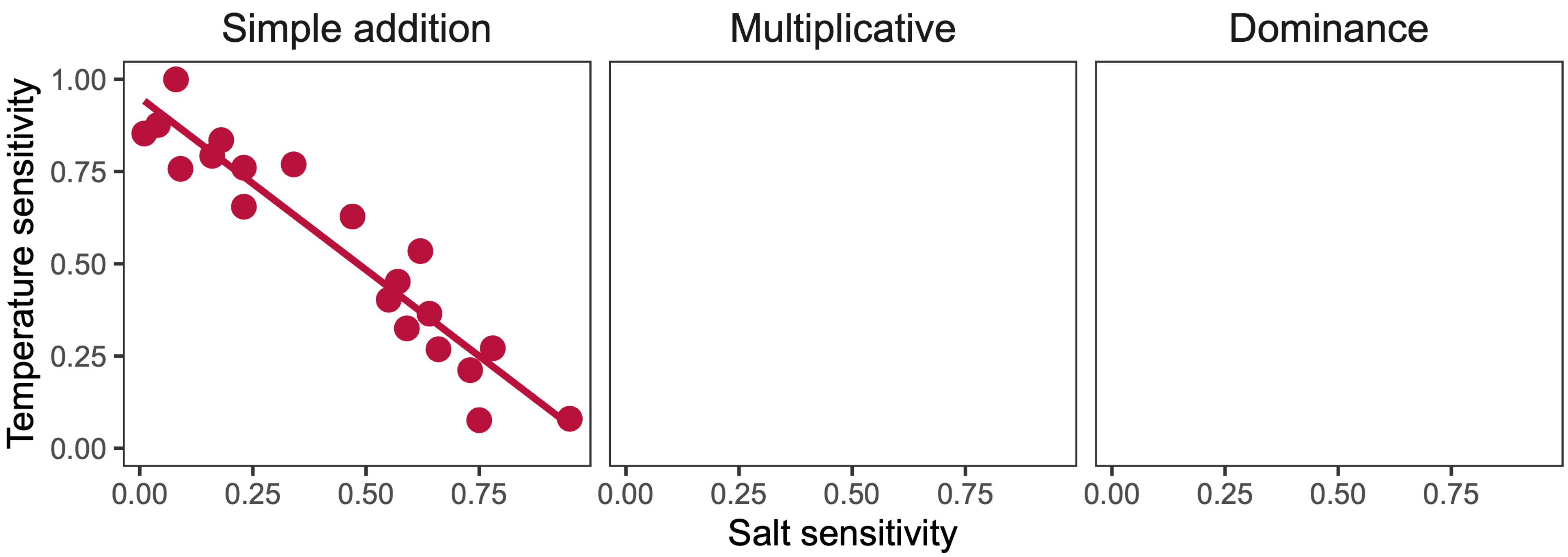
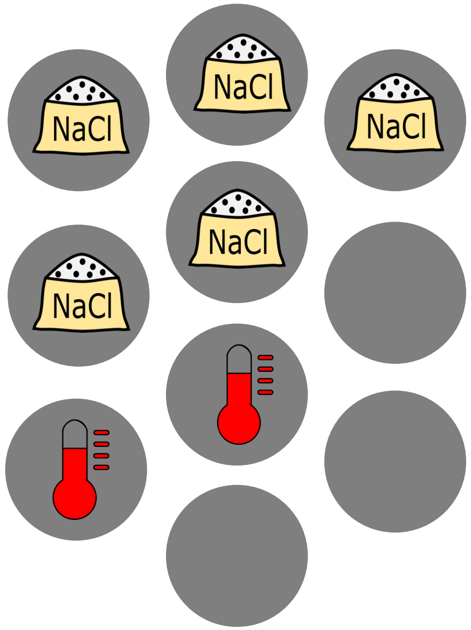
Stressor sensitivity correlation





Negative correlation:
Each stressor affects a different set of individuals (or species)

Madge Pimentel et al. (2025). In: *Ecology & Evolution*

Which null model should I choose?

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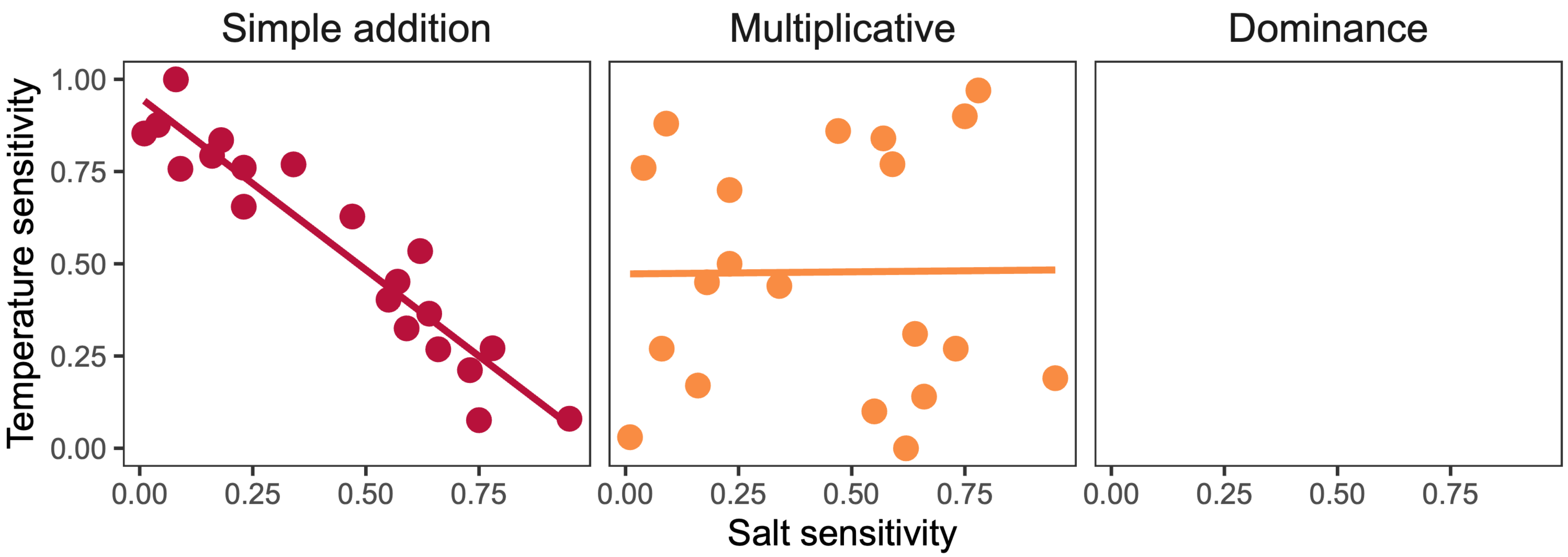
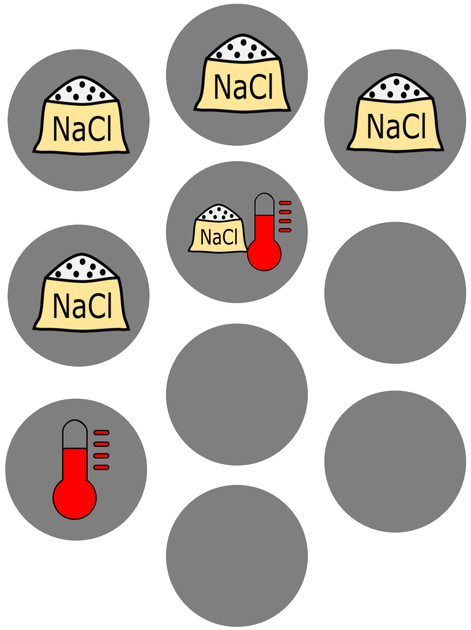
Stressor sensitivity correlation





Negative correlation:
Each stressor affects a different set of individuals (or species)

No correlation:
Stressor sensitivities are independent of each other

Madge Pimentel et al. (2025). In: *Ecology & Evolution*

Which null model should I choose?

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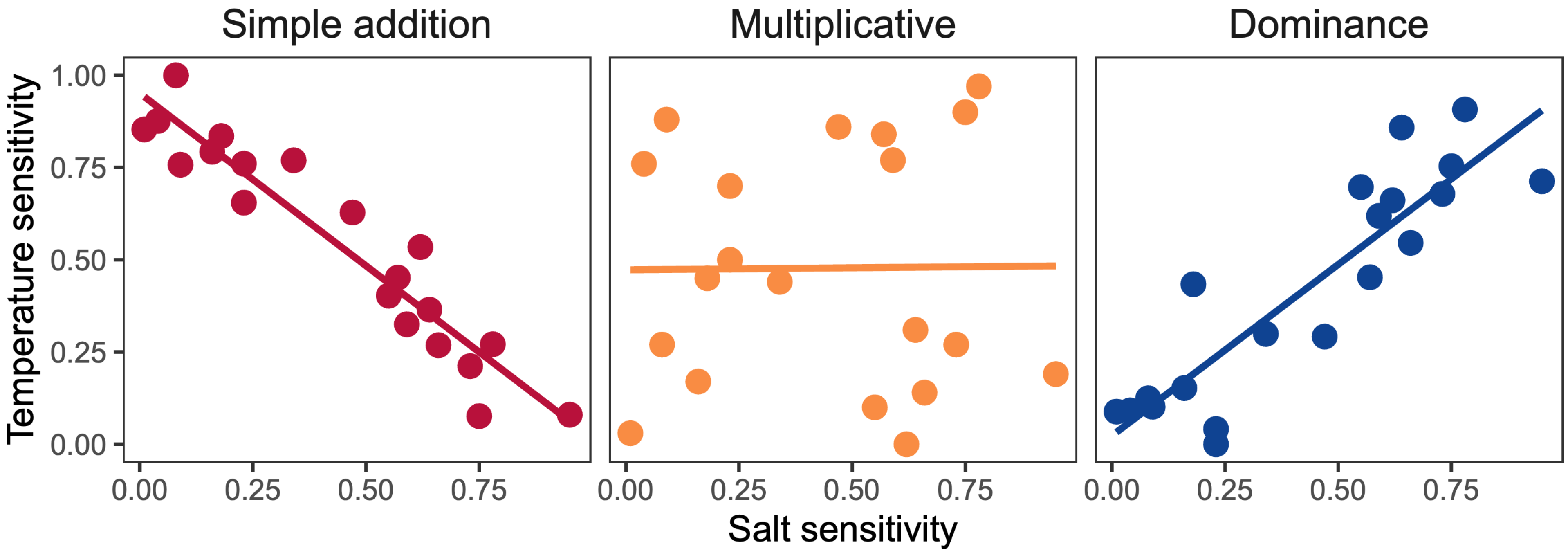
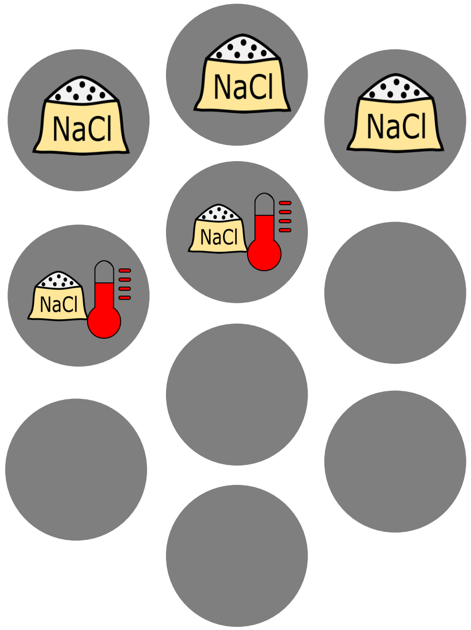
Stressor sensitivity correlation





Negative correlation:
Each stressor affects a different set of individuals (or species)

No correlation:
Stressor sensitivities are independent of each other

Positive correlation:
If an individual (or species) is affected by one stressor, it is likely to be also affected by the other

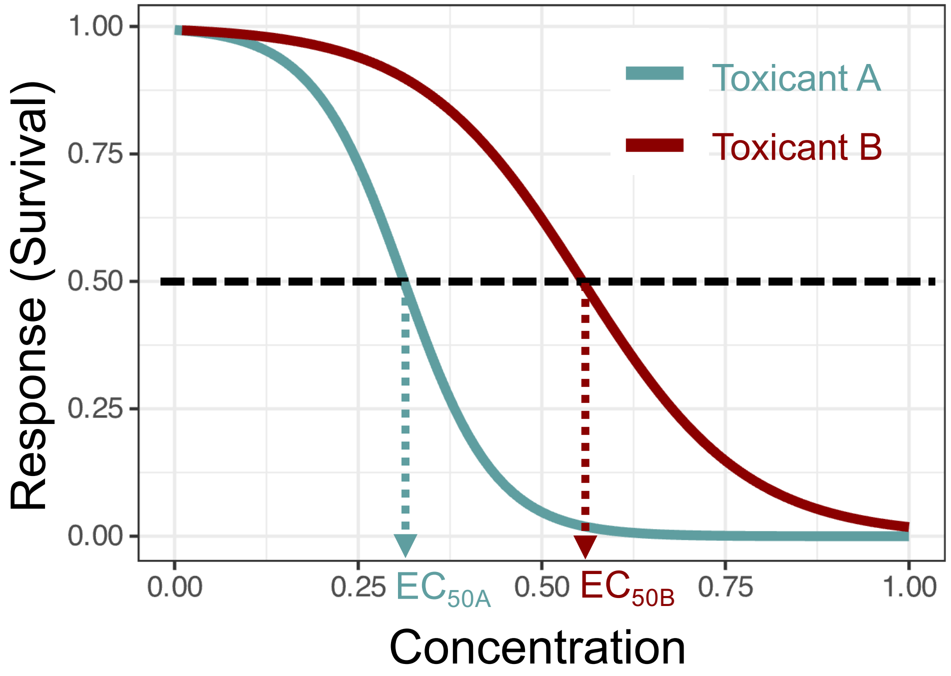
Madge Pimentel et al. (2025). In: *Ecology & Evolution*

Toxicant mixtures and mode of action: Null models in ecotoxicology

Similar modes of action: Concentration addition

$$f_{A,B}(c_A, c_B) = f_A(c_A + \lambda c_B)$$

$$\text{Potency } \lambda = \frac{EC_{xA}}{EC_{xB}}$$



- Both toxicants are in principle **interchangeable**
- We can translate the concentration of toxicant B to an equivalent concentration of toxicant A by **scaling with its potency**
- We can also extend CA to stressors, replacing concentrations with stressor intensity
  - Then, we assume both stressors to have similar physiological effect chains

Toxicant mixtures and mode of action: Null models in ecotoxicology

Different modes of action: Independent action

$$E_{A,B} = E_A + E_B - E_A E_B$$

This formulation of independent action refers to the effect of the toxicant/stressor: The mortality caused by it

$\downarrow$ $E_X = 1 - \frac{S_X}{C}$

$$1 - \frac{S_{A,B}}{C} = 1 - \frac{S_A}{C} + 1 - \frac{S_B}{C} - \left(1 - \frac{S_A}{C}\right)\left(1 - \frac{S_B}{C}\right)$$

$\downarrow$ Simplification

This formulation for the multiplicative null model requires the response variable to increase with better conditions (i.e., it works with survival as response, but not with mortality)

$$S_{A,B} = \frac{S_A S_B}{C}$$

Independent action is equivalent to the multiplicative null model!

Multiple-stressor and chemical mixture research can complement each other

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Multiple stressor research	Joint challenges & research gaps	Chemical mixture research
tools and methods		tools and methods
♦ Significance testing ♦ Null models integrating communities developed	Too focused on interactions Null model selection	♦ Concept of Model Deviation Ratio ♦ Null models integrating chemical & non-chemical stressors ♦ Large databases on chemical effects

Simple addition, multiplicative, dominance, compositional null model

Concentration addition, independent action, stress addition null model

Schäfer et al. (2023). In: *Environmental Toxicology and Chemistry*

Stress addition model for mortality: Combining environmental and toxicant stressors?

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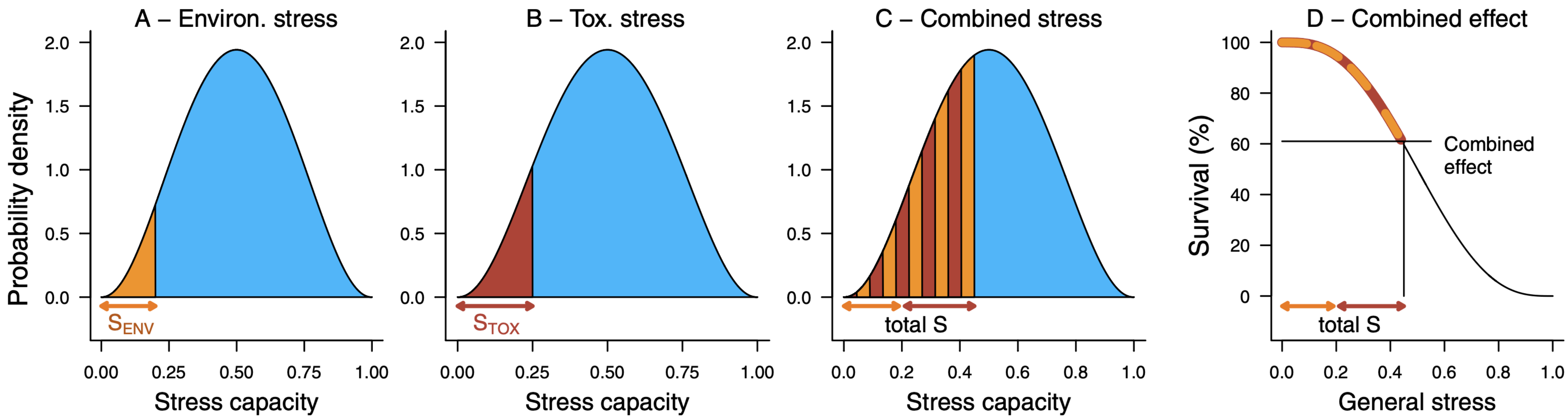

Environmental stressor and toxicant act on a general stress capacity. Individuals have low or high stress capacities and die as soon as their capacity is exceeded. SAM generally predicts higher mortality than the other null models.

Liess et al. (2016). In: *Scientific Reports*

So – How do we choose how to predict joint effects?

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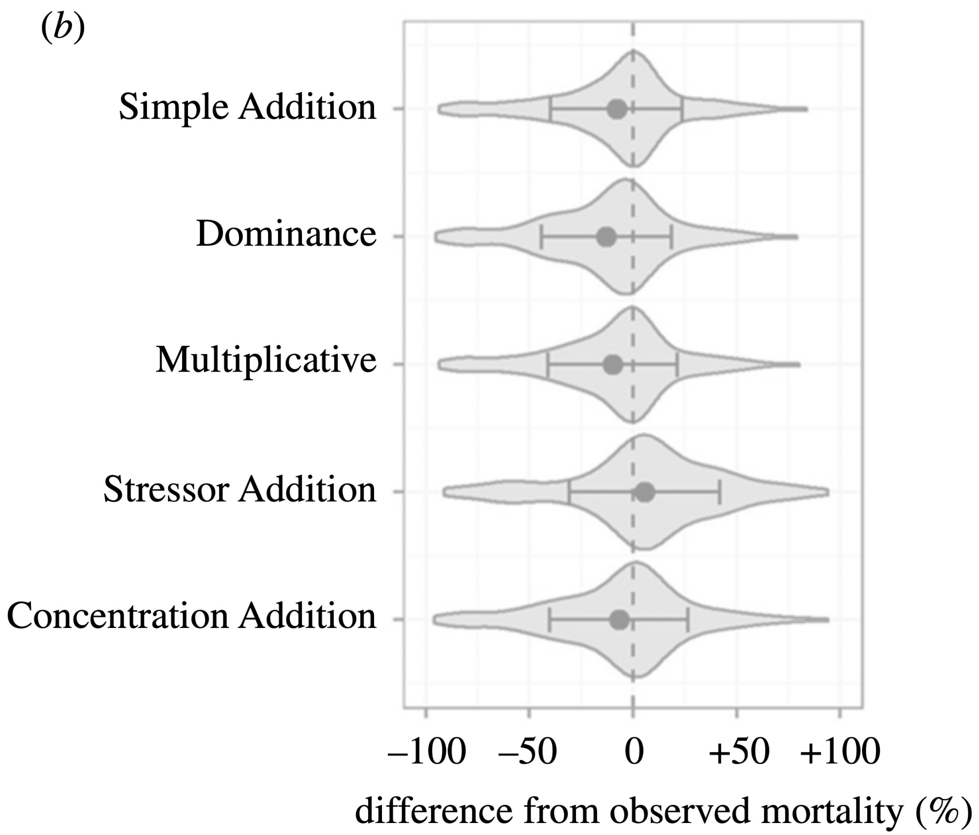
- Stressors **mode of action**
  - How do the stressors affect the organisms?
    - physically, specific or non-specific physiologically
  - Similar or different mode of action?
  - Are stressor interactions known?
- **Sensitivities**: Relationship between sensitivities to stressors?
- **Response** variable: What is affected?
  - e.g., survival rather multiplicative, growth rather additive
- **Effect size**: What are the individual effect sizes?
  - e.g., close to limit or differs by orders of magnitude: dominance
- Knowledge of **stressor-effect relationship**: Broader range of models available

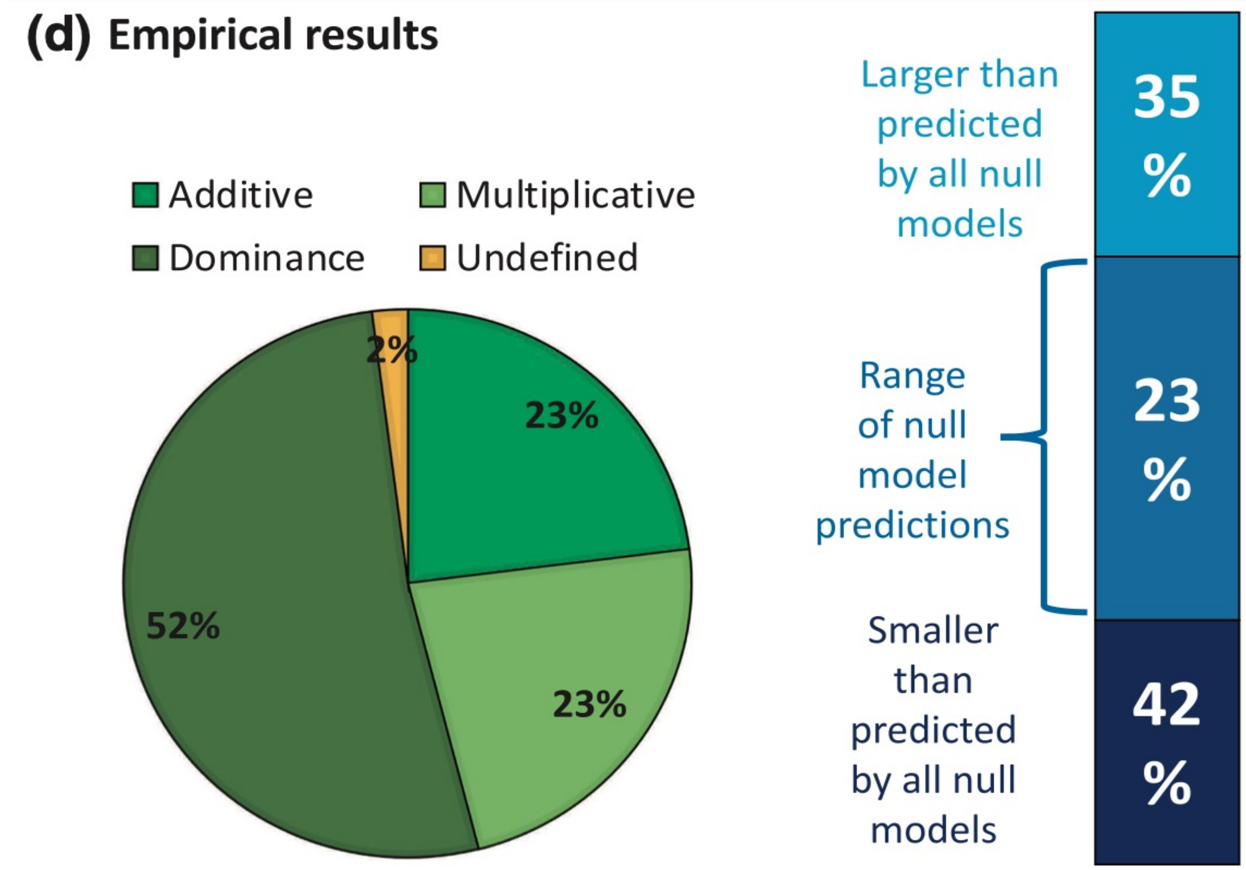
Comparing null models: Mortality as effect

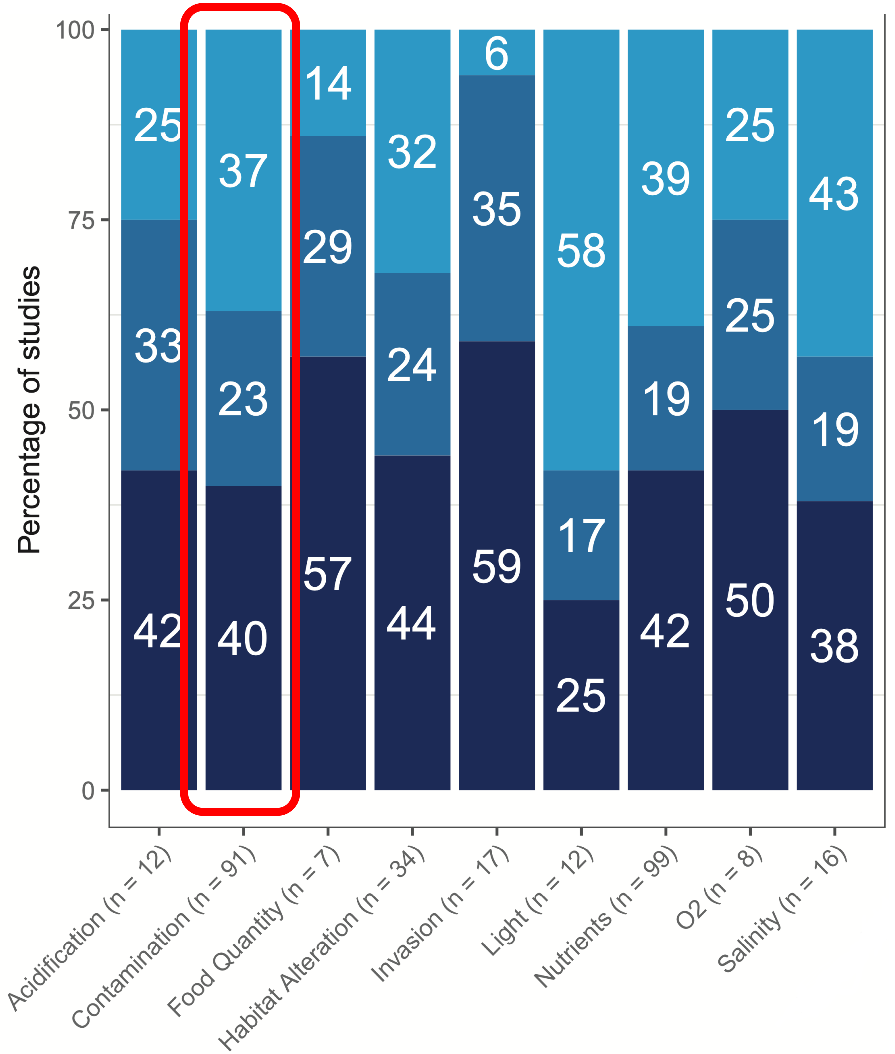
SAM (and Concentration Addition model) least biased in meta-analysis (but: only mortality as effect)



Dey & Koops (2021). In: *Proceedings of the Royal Society B: Biological Sciences*

Comparing null models: Joint stressor effects often outside of range of multiple-stressor null models





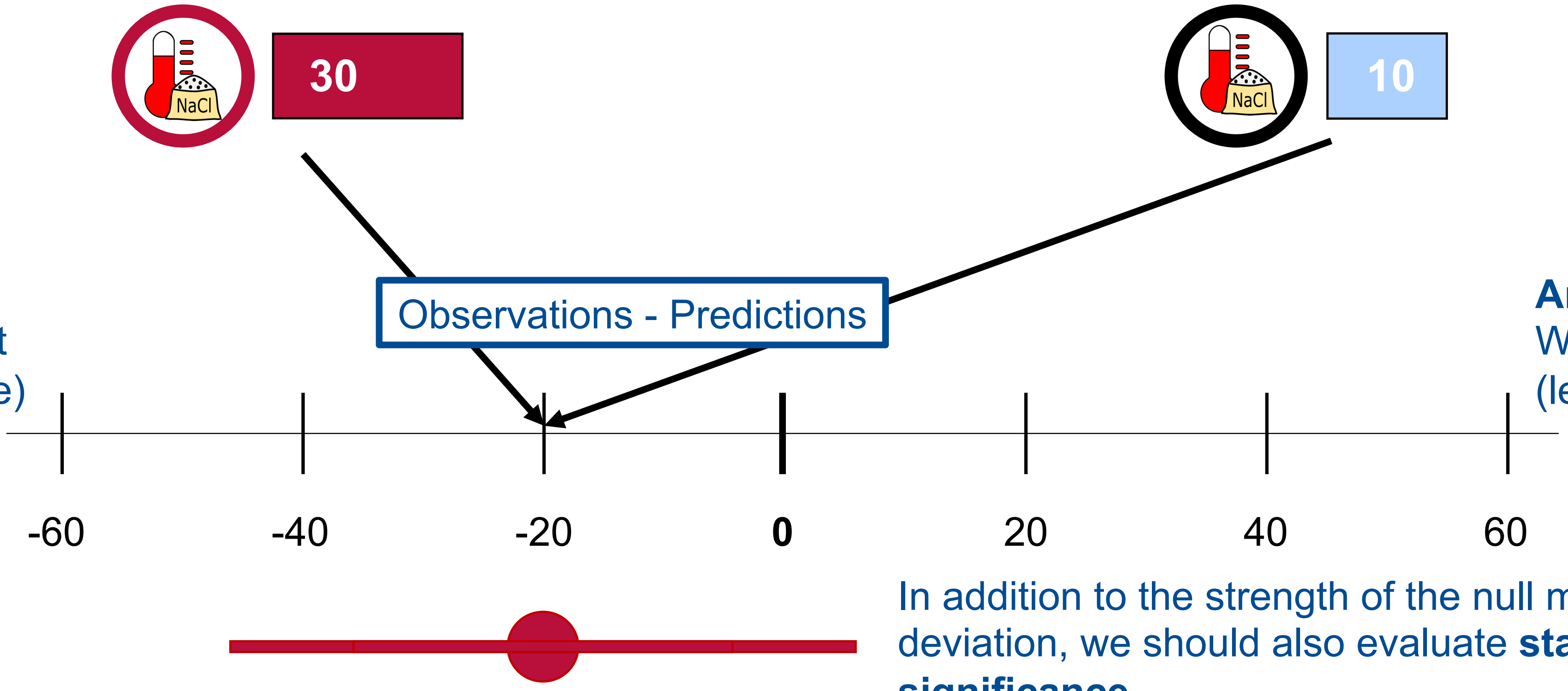
Dominance null model definition different from the one given before!

Morris et al. (2022). In: *Ecology Letters*

How can we test for stressor interactions?

Null model predictions
(Simple addition)

Experimental observations



Synergism:
Stronger effect
(more negative)

Antagonism:
Weaker effect
(less negative)

In addition to the strength of the null model deviation, we should also evaluate **statistical significance**

Interaction terms in multiple linear regression

$$y = \boxed{\beta_0} + \boxed{\beta_1 x_1} + \boxed{\beta_2 x_2} + \boxed{\beta_3 x_1 x_2}$$

Intercept

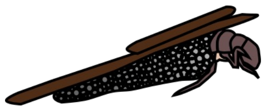
What is the response if both predictors are absent (at value 0)?

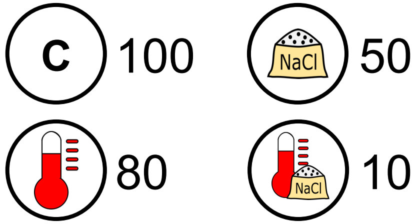
Predictor main effects

How much does the response change for one unit increase in x_1 (or x_2)?

Interaction effect

How much does the per-unit effect of x_1 change with one unit increase in x_2 (and vice versa)?



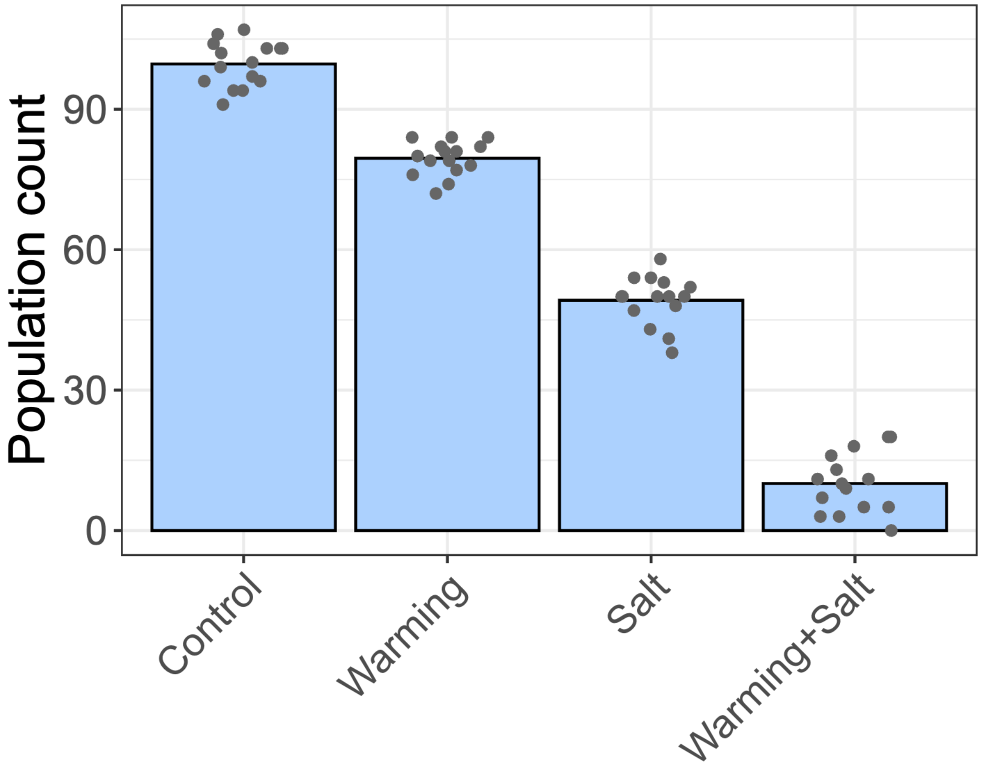


$$y = 100 - 20x_1 - 50x_2 - 20x_1 x_2$$

Linear regression evaluates the simple addition null model

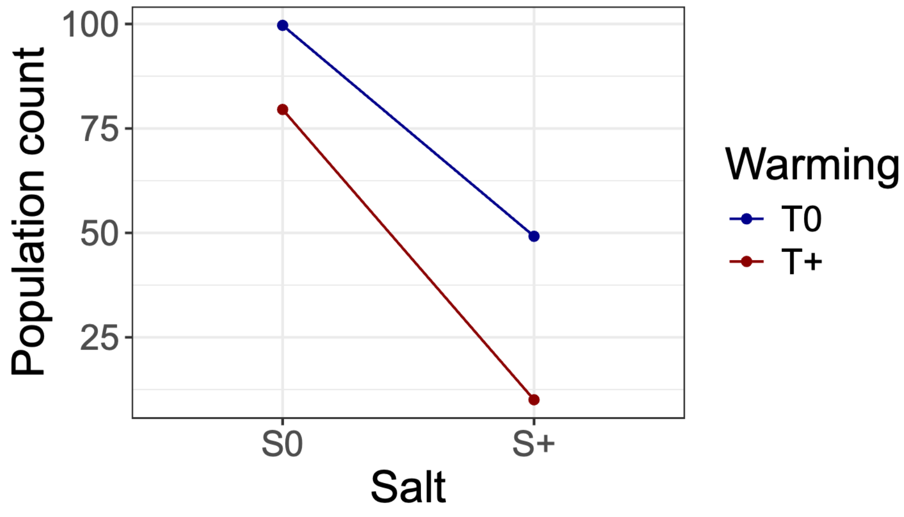
Interactions between categorical predictors

By how much does the effect of salt change if warming is also present? (And vice versa)



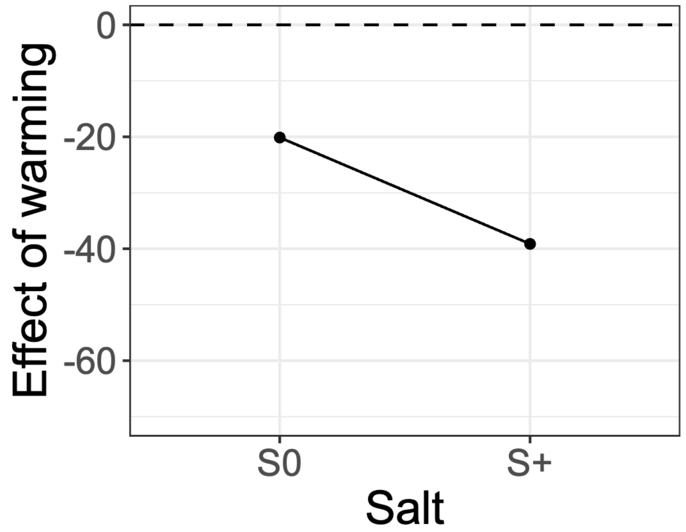
Null model:
Parallel lines

Conditional plot

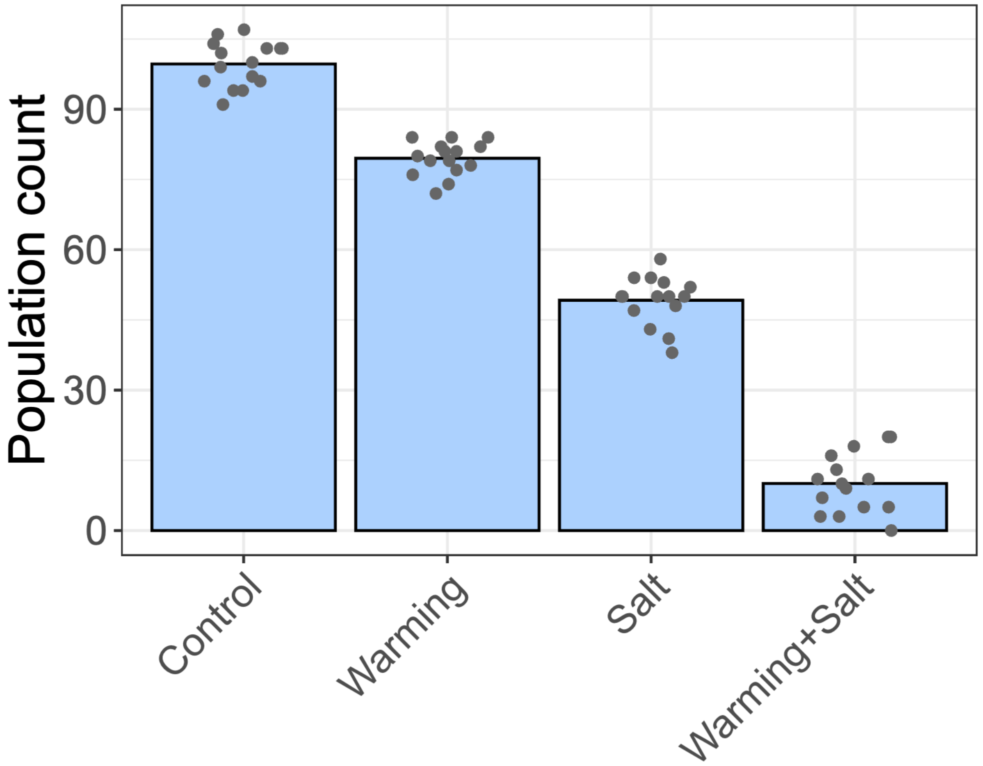


Marginal effects plot

Null model:
Horizontal line



Interactions between categorical predictors



Null model:
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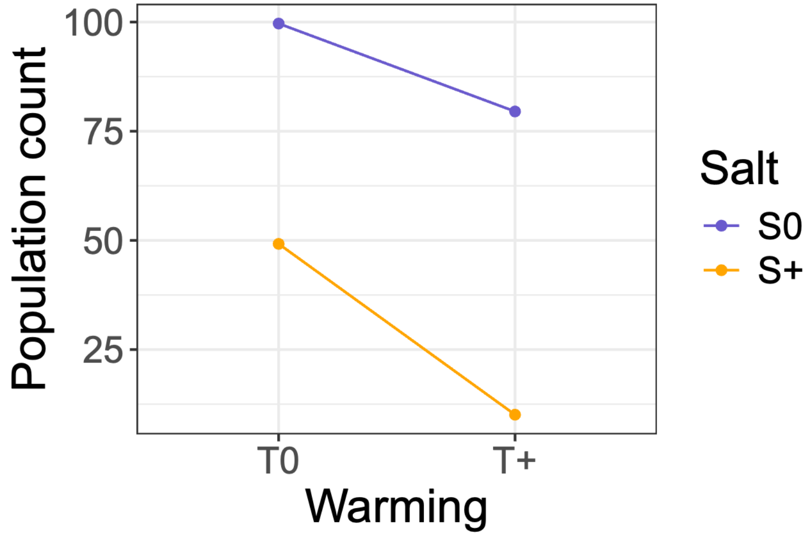
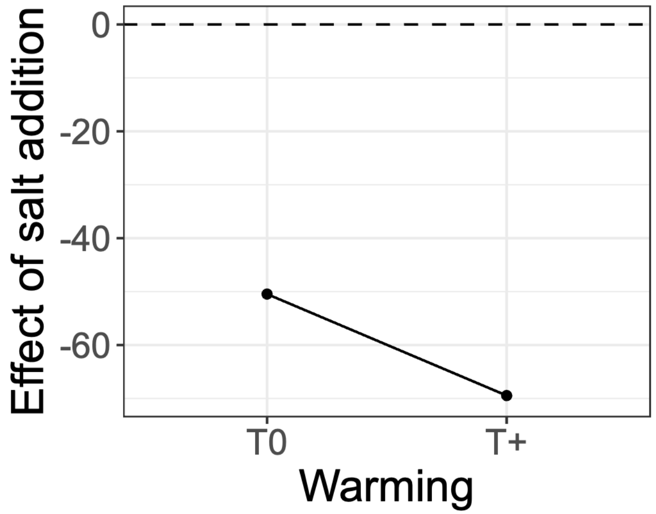
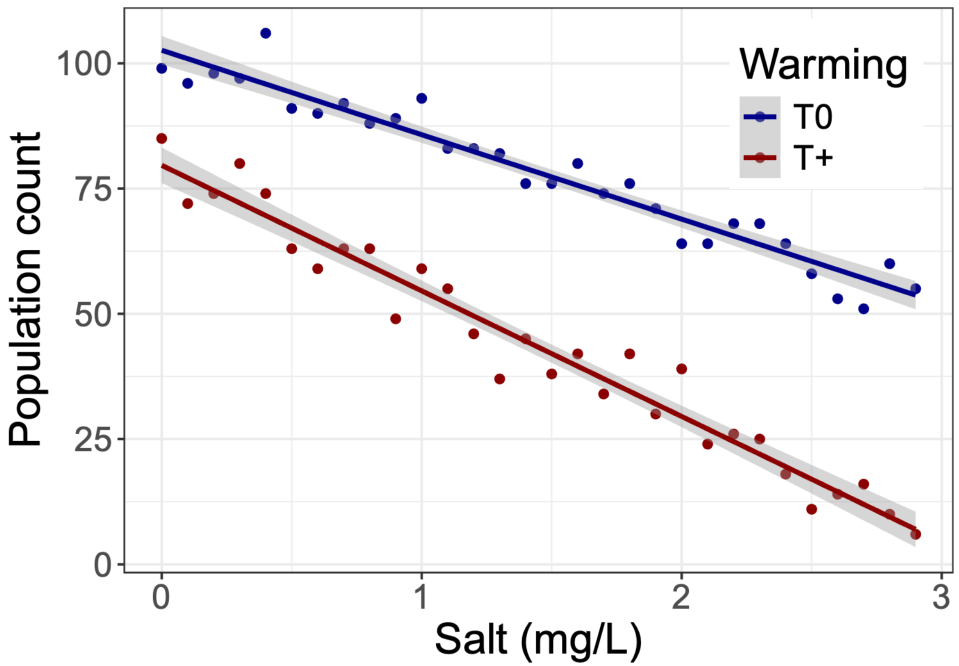
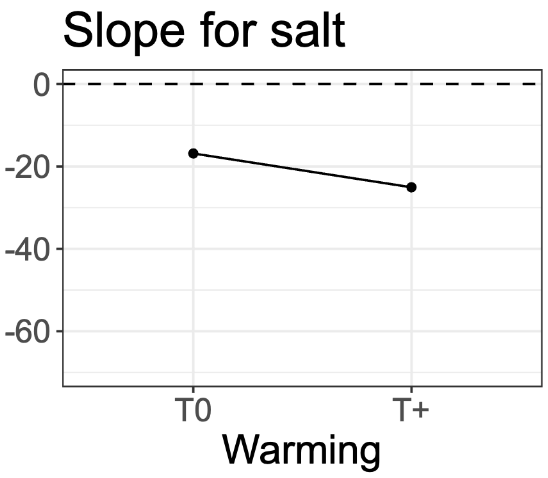




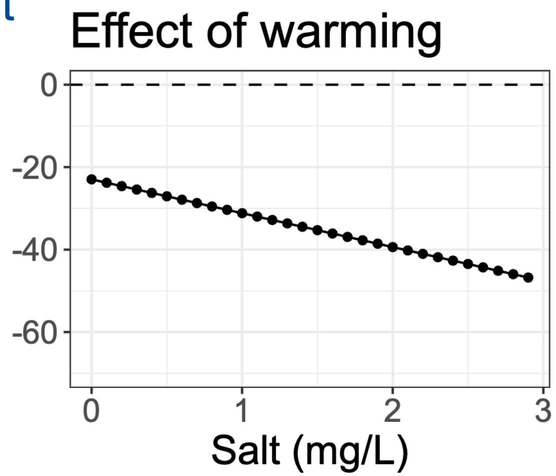
Interaction between a categorical and continuous predictor



Marginal effects plot

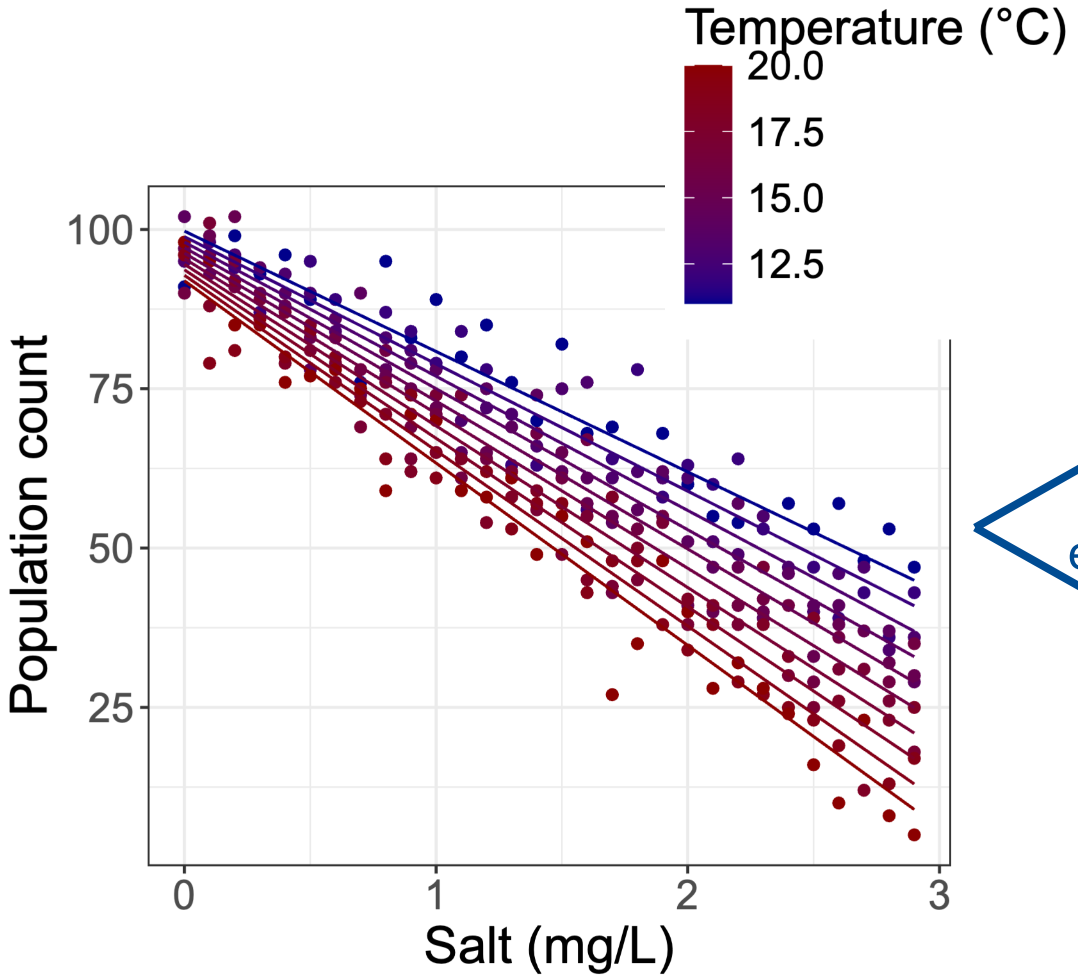


Describes how the slope of the salt regression line changes between the two temperature conditions

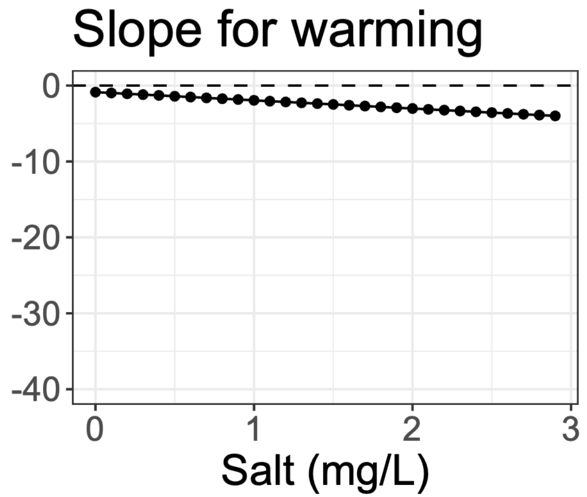


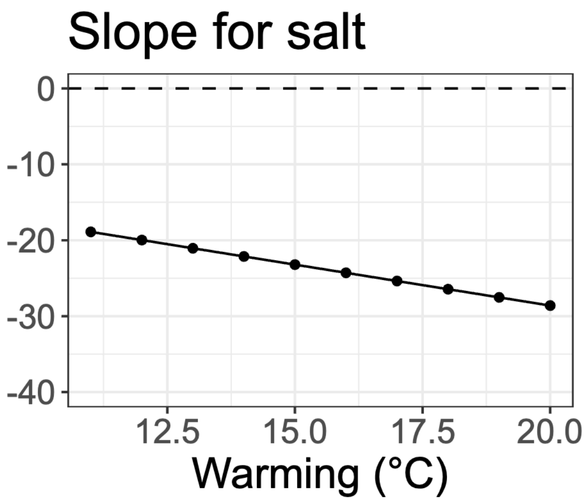
Describes how the difference in population size between control and warming changes with increasing salinity

Interaction between two continuous predictors



Marginal effects plot





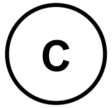
Describes how the slope of the regression line for stressor A changes for a unit increase in stressor B

Interpretation of model output

2 categorical predictors

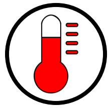
```
Coefficients:
                         Estimate Std. Error
(Intercept)                99.667      1.321
temperatureT+             -20.133      1.869
salinityS+                -50.467      1.869
temperatureT+:salinityS+  -19.000      2.643
---
```

$$y = 99.7 - 20.1 \times T - 50.5 \times S - 19 \times T \times S$$

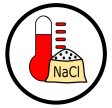


$T = 0, S = 0$

$T = 0, S = 1$



$T = 1, S = 0$



$T = 1, S = 1$

2 continuous predictors

```
Coefficients:
                      Estimate Std. Error
(Intercept)           108.6765     3.2143
temperature            -0.8525     0.2039
salinity               -7.2802     1.9034
temperature:salinity   -1.0395     0.1207
---
```

$$y = 108.7 - \frac{0.9 \times T}{°C} - \frac{7.3 \times S}{mg/L} - \frac{1 \times T \times S}{°C \times mg/L}$$

Slope of temperature for 0 mg/L of salt

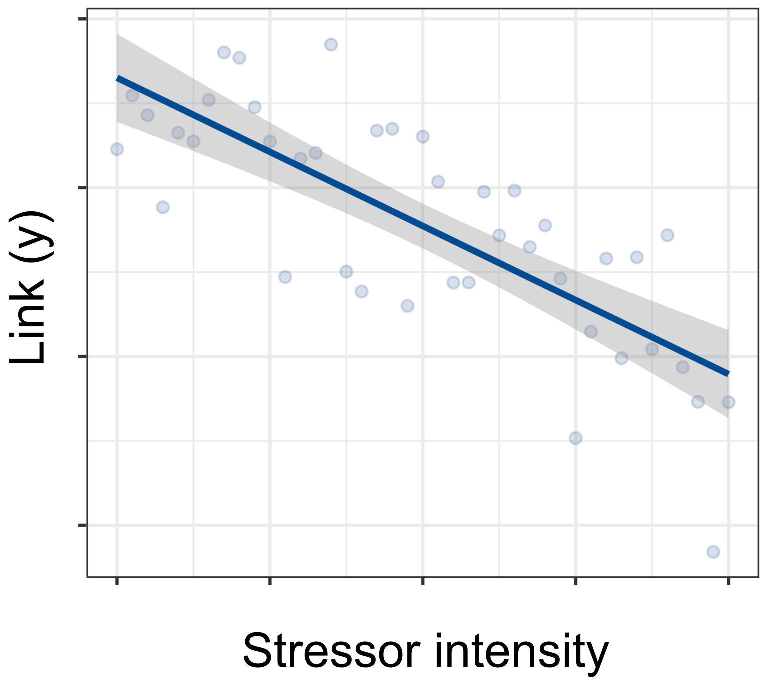
Slope of salinity for 0 °C

Interactions in Generalized linear models: Link scale vs. Response scale

Response variable and link function

$$\boxed{\mathrm{link}(y)} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

Link function linearizes stressor-response relationships



Identity link

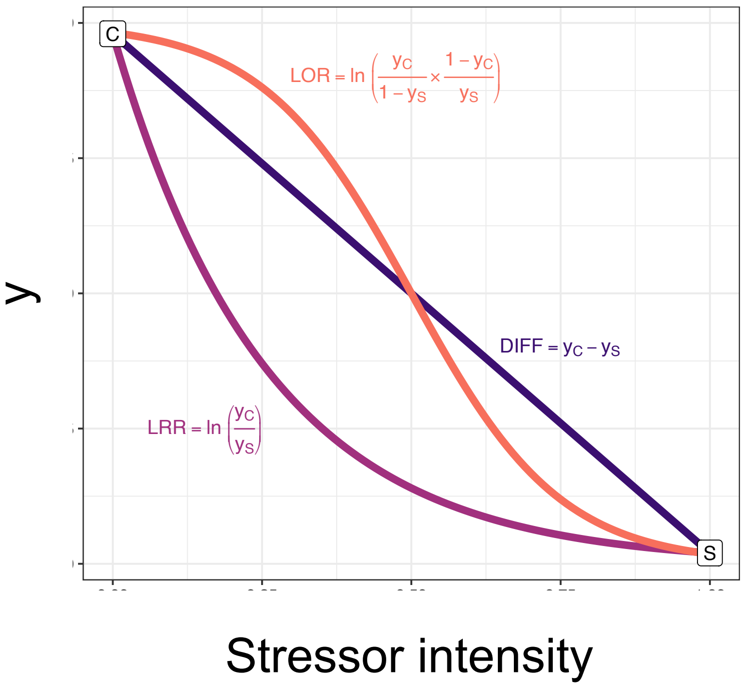
(Gaussian regression)

Logarithmic link

(Poisson regression)

Logit link

(Binomial regression)



GLMs have a canonical link function

GLM	Used for…	Limits of response	Canonical link
Gaussian	Normal data	$(-\infty, +\infty)$	Identity
Poisson	Count data	$(0, +\infty)$	Logarithmic
Binomial	Proportional data	$(0, 1)$	Logit
Gamma	Continuous data ≥ 0	$(0, +\infty)$	Inverse

- Canonical link function facilitates the estimation process for the model parameters
- Most typically used link function for a GLM
- Other link functions can also be used, but this may lead to convergence issues

What does the (canonical) link function do?

It determines…

1. **The stressor-response relationship that is modelled**
2. **The effect measure**
   - Differences (identity link)
   - Ratios (logarithmic link)
   - Odds ratios (logit link)
3. **How the predictors’ effects are expected to combine in the absence of interactions**

How the logarithmic link implies the multiplicative null model

Absence of interaction: $\beta_3 = 0$

Apply inverse link function (here: exponentiate)

Separate predictor effects according to algebraic rules

$$\log(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

$$\downarrow$$

$$\log(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$\downarrow$$

$$y = e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2}$$

$$\downarrow$$

$$y = e^{\beta_0} \times e^{\beta_1 x_1} \times e^{\beta_2 x_2}$$

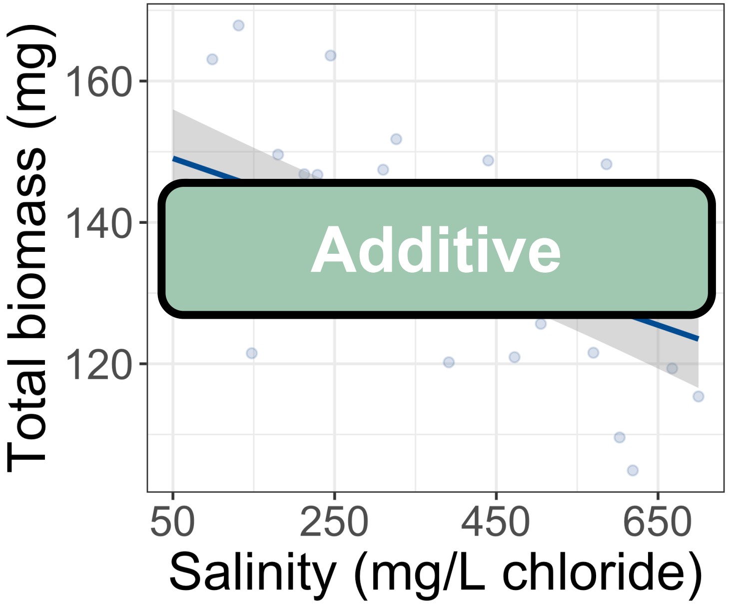
The logarithmic link function in a GLM evaluates the multiplicative null model

Multiple regression and interaction terms

$$link(y) = \alpha + \beta_S \times Salt + \beta_F \times Flow + \boxed{\beta_{Int} \times Salt \times Flow}$$

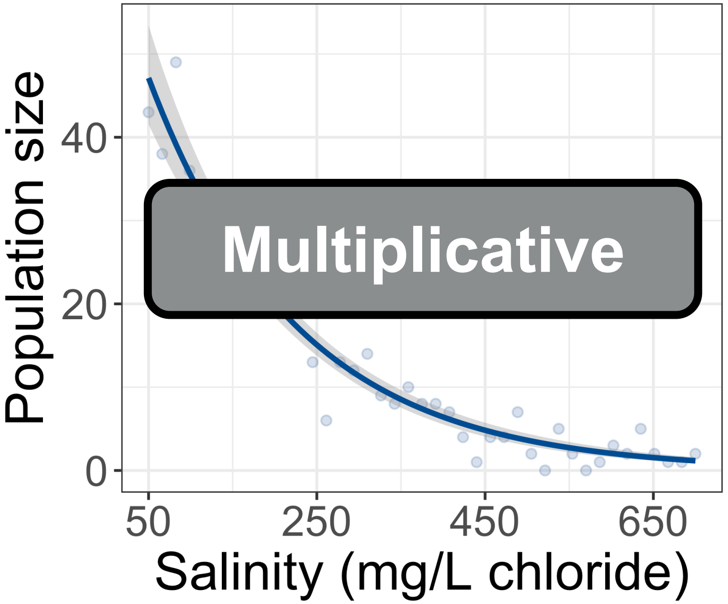
Gaussian regression

Identity link



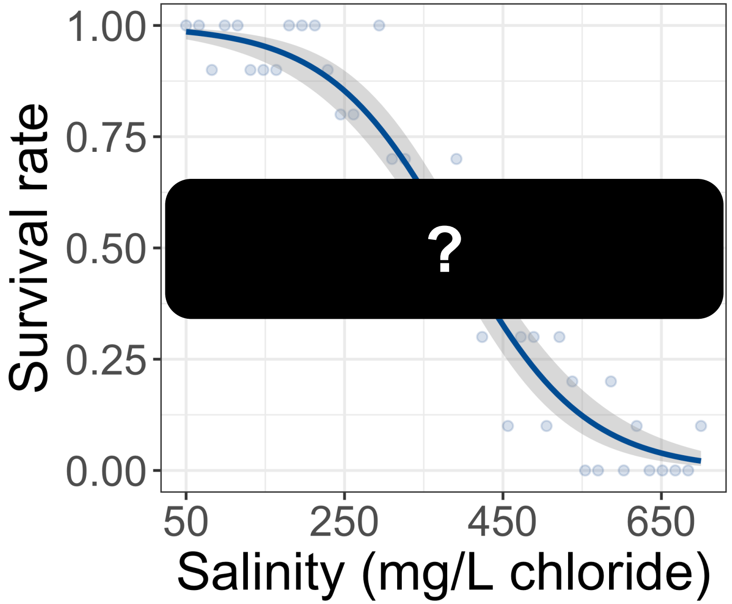
Poisson regression

Logarithmic link



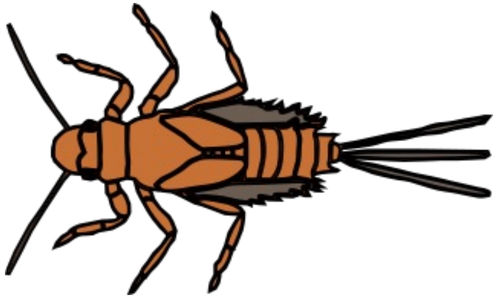
Binomial regression

Logistic link



Case-study: Interaction of reduced flow velocity and salinization on *Baetis* sp.?

(1) Choose multiple-stressor null model



Toxicity, costs for osmoregulation

Habitat and oxygen availability

Distinct modes of action & negatively correlated sensitivities

Additive null model

Baetis sp.: © Mandy Sander

Case study: Beermann et al. (2018). In: *Science of the Total Environment, 610*

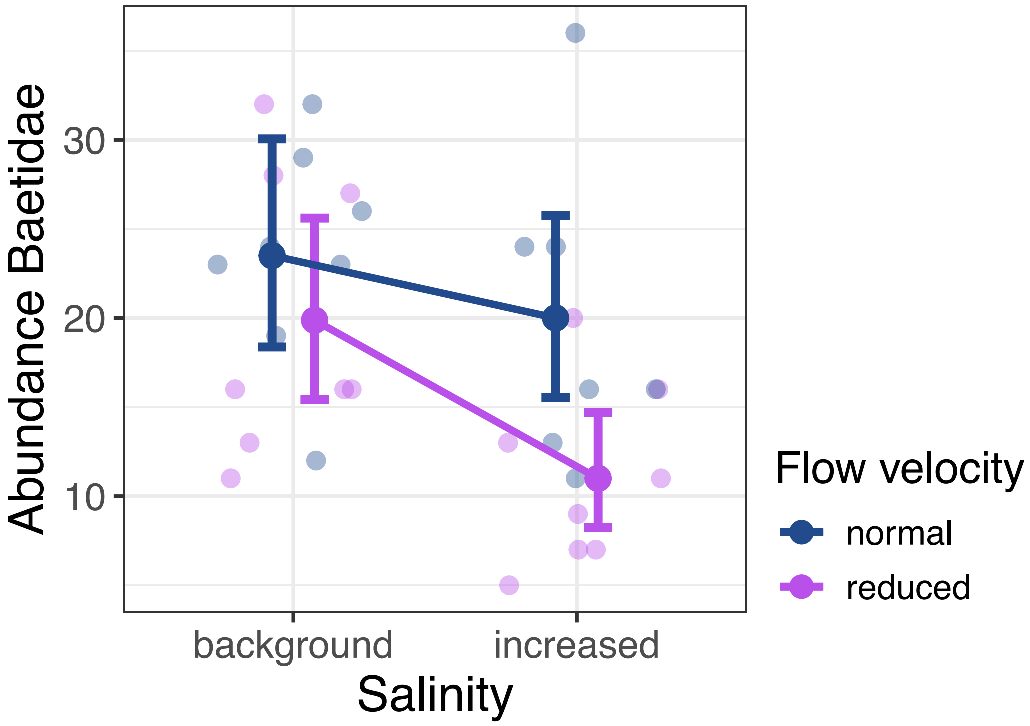
Case-study: Interaction of reduced flow velocity and salinization on *Baetis* sp.?

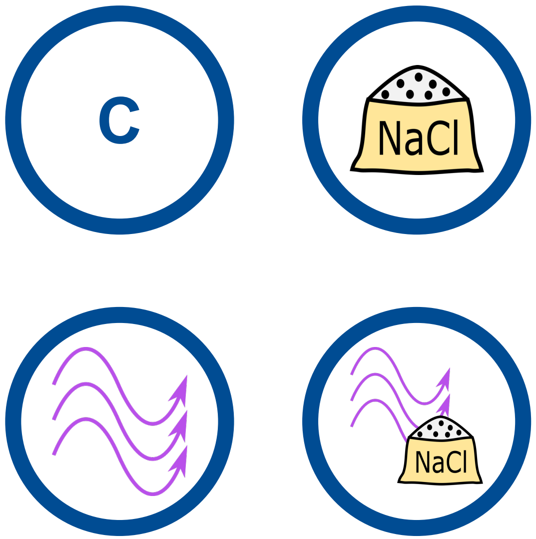
(1) Choose multiple-stressor null model

(2) Statistical model specification

Poisson regression (log-link):

$$log(Abund) = \alpha + \beta_F \times F + \beta_S \times S + \beta_{Int} \times F \times S$$





Data from Beermann et al. (2018). In: *Science of the Total Environment, 610*

Post-estimation hypothesis testing (PEHT)

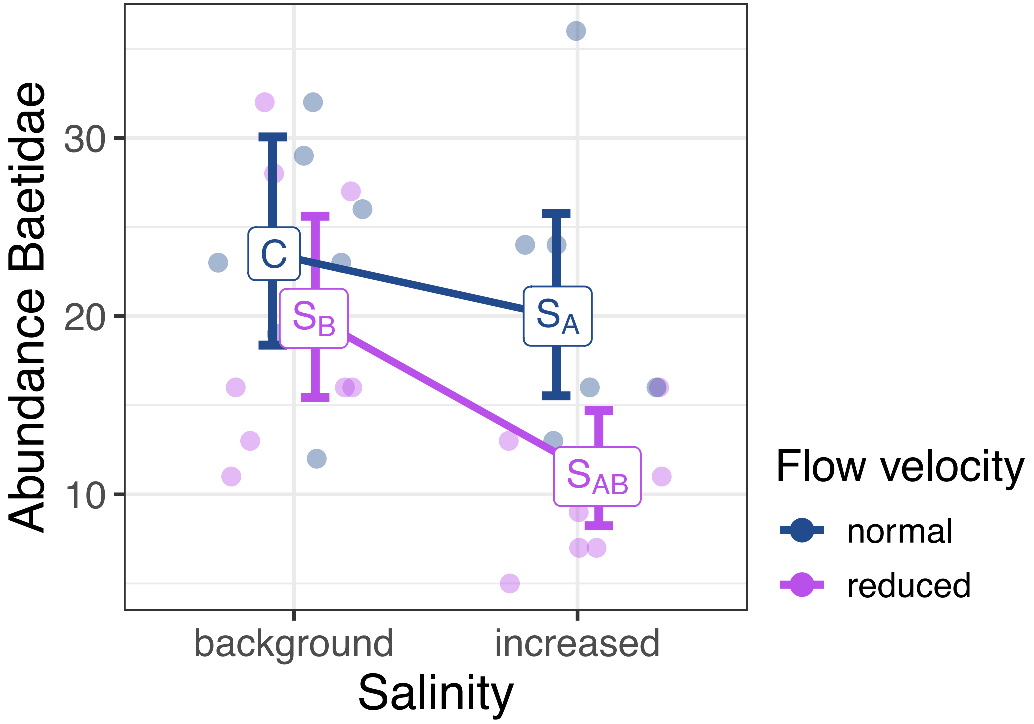
(1) Choose multiple-stressor null model

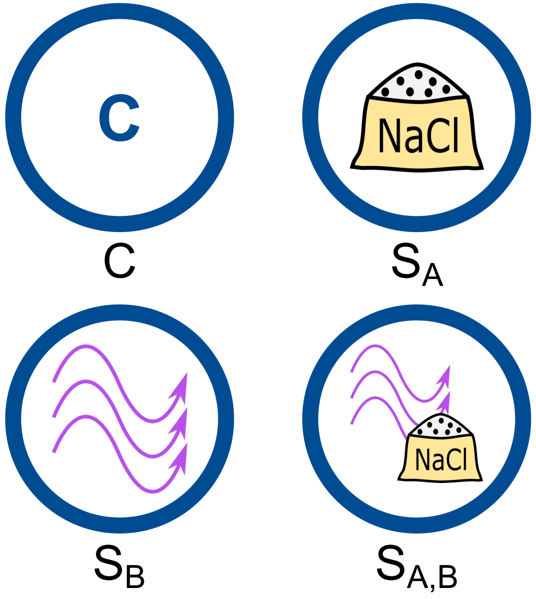
(2) Statistical model specification

(3) Hypothesis-tests on stressor interactions

Additive null model: $S_{A,B} = S_A + S_B - C$

Interaction estimate: $int_{AD} := S_{A,B} - S_A - S_B + C$





Data from Beermann et al. (2018). In: *Science of the Total Environment, 610*

Calculation of interaction estimate and its uncertainty

Additive

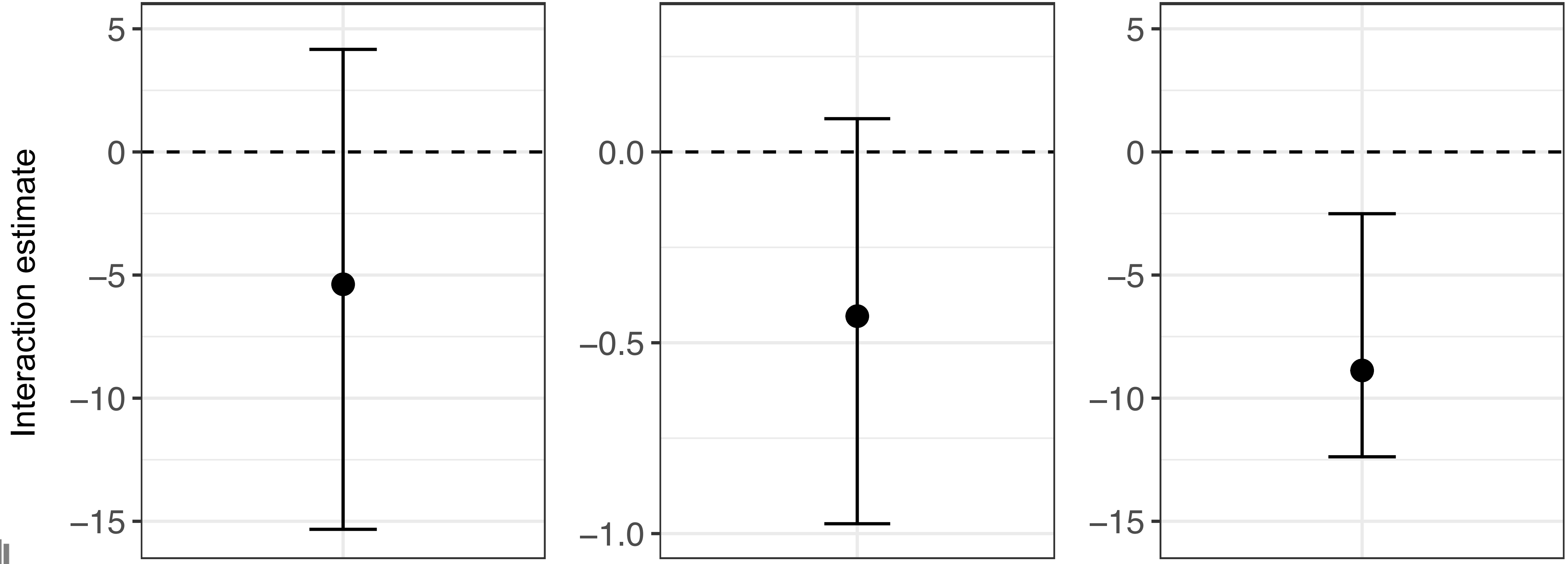
$$S_{A,B} = S_A + S_B - C$$

Multiplicative

$$S_{A,B} = \frac{S_A \times S_B}{C}$$

Dominance

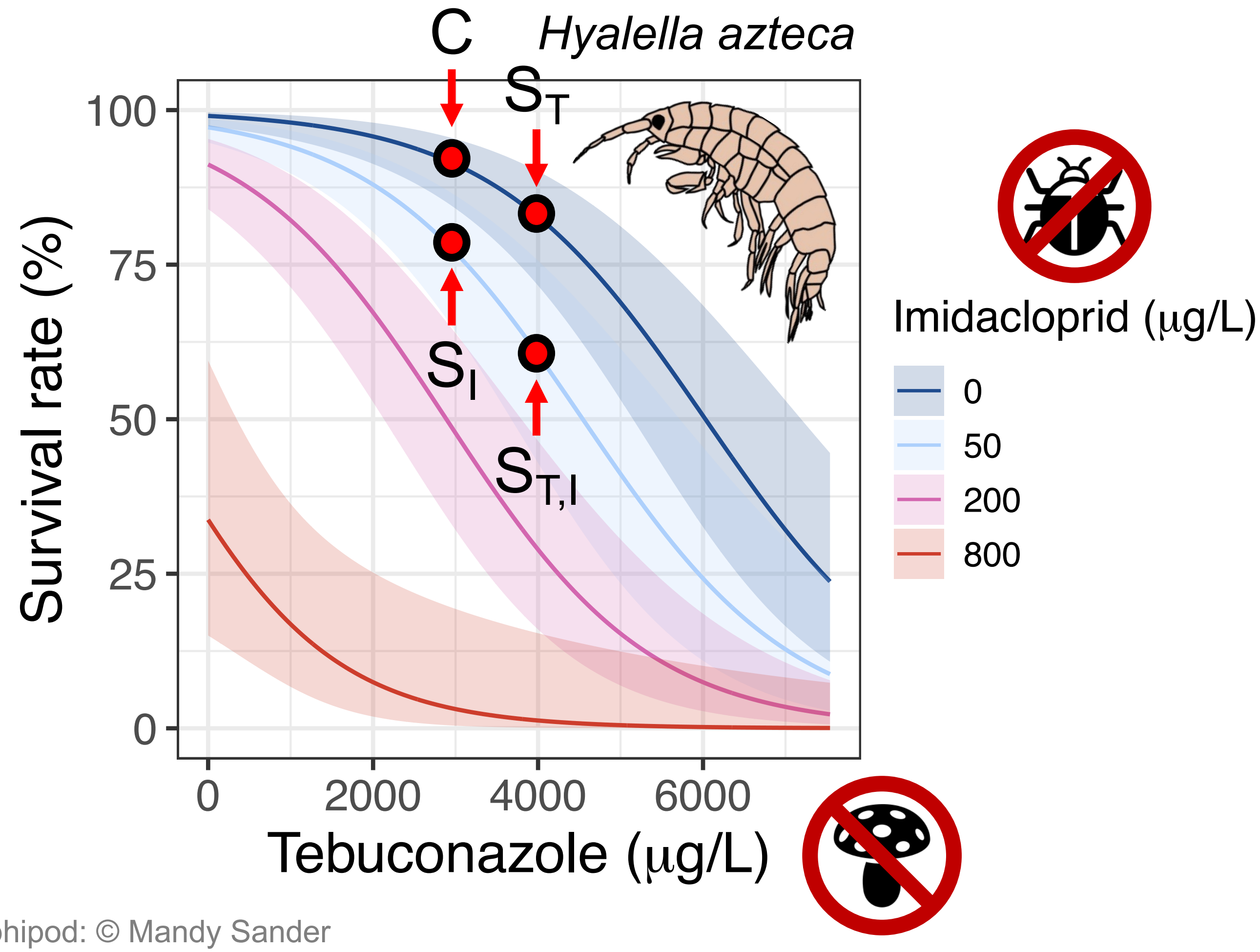
$$\min\left(S_{A,B}, C\right) = \min(S_A, S_B)$$



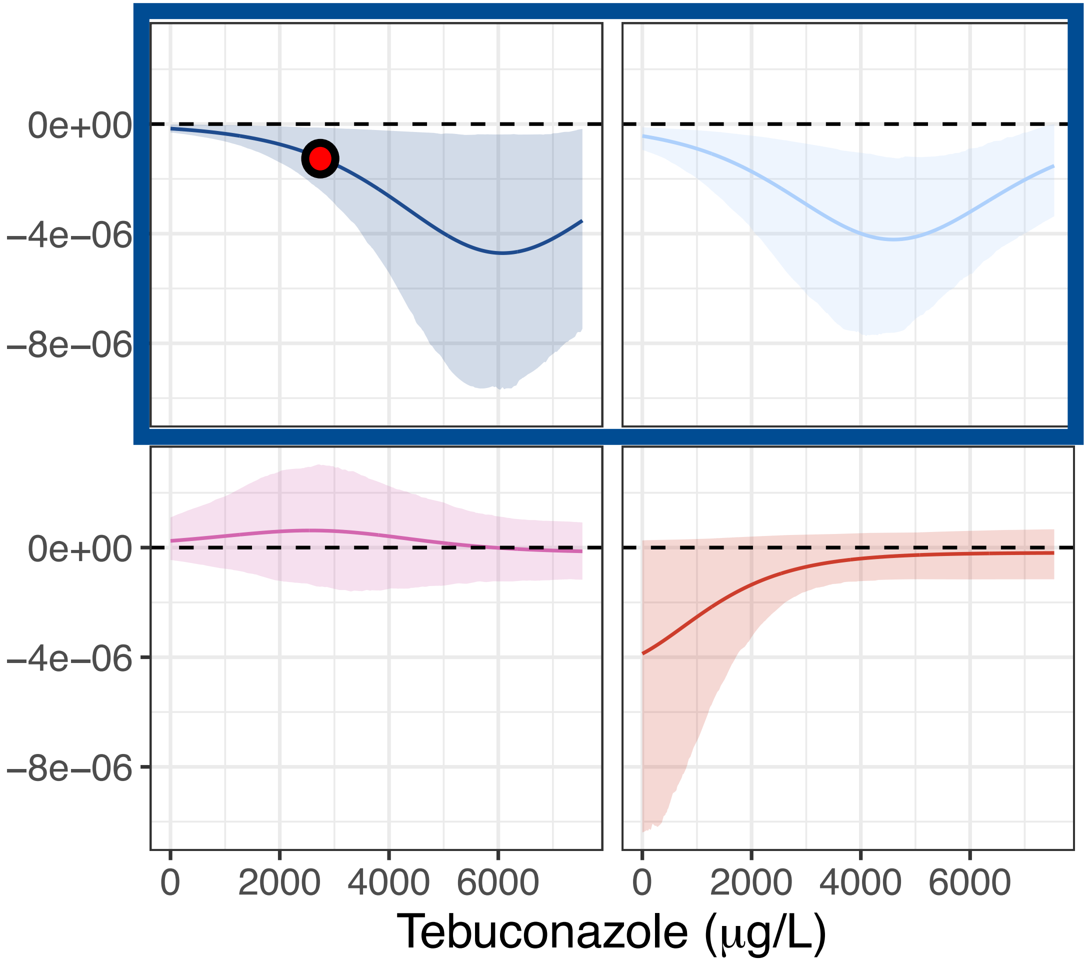
Madge Pimentel et al. (2025). In: *Ecology & Evolution*

Extension to gradient designs

Insecticide x Fungicide



Interaction estimate (Multiplicative)



Amphipod: © Mandy Sander
Based on data from Raby et al. (2019). In: *Environmental Toxicology and Chemistry, 38*

Summary

1. **Stressor interactions are deviations from null model predictions**
2. **Null model choice should be based on *a priori* knowledge on…**
   - Similarity in stressor/toxicant mode of action
   - Sensitivity correlations
3. **GLMs make their own assumption on how the effects of predictors combine in the absence of an interaction**
   - Identity link: Simple addition null model
   - Logarithmic link: Multiplicative null model
4. **But: We can use post-estimation hypothesis testing to evaluate any of the null models based on the same regression model!**